



MA 606: Mathematical Models for Infectious Diseases

Course Code	MA 606
Course Name	Mathematical Models for Infectious Diseases
L-T-P-C	3-0-0-3
One-time approval	26th BoA (Oct 2018); 18th Senate (Oct 2018)

Course Contents

Over the last two decades, mathematical models have seen a huge development in all aspects of infectious diseases, from microbiology to epidemiology and evolution. The programme covers introductory and advanced concepts in mathematical modelling of infectious diseases, including:

- Mathematical review (calculus, probabilities ...)
- Deterministic and stochastic models
- Network analysis
- Within-host dynamics of viral and bacterial infections
- Applied programming with R
- Statistical data modelling
- Computer-based simulations